

Volume 4 — Wiring Diagrams & I/O Reference

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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⚠ **STATUS:** Diagrams and tables reflect the **current pin authority** (planning stage). Rows marked **PLAN_VERIFY** / **ACCEPTED_VERIFY** must be traced in the cabinet before landing wires. The original Mazak schematic set **41434WB** remains authoritative for machine-side wiring.

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§1 — System Architecture

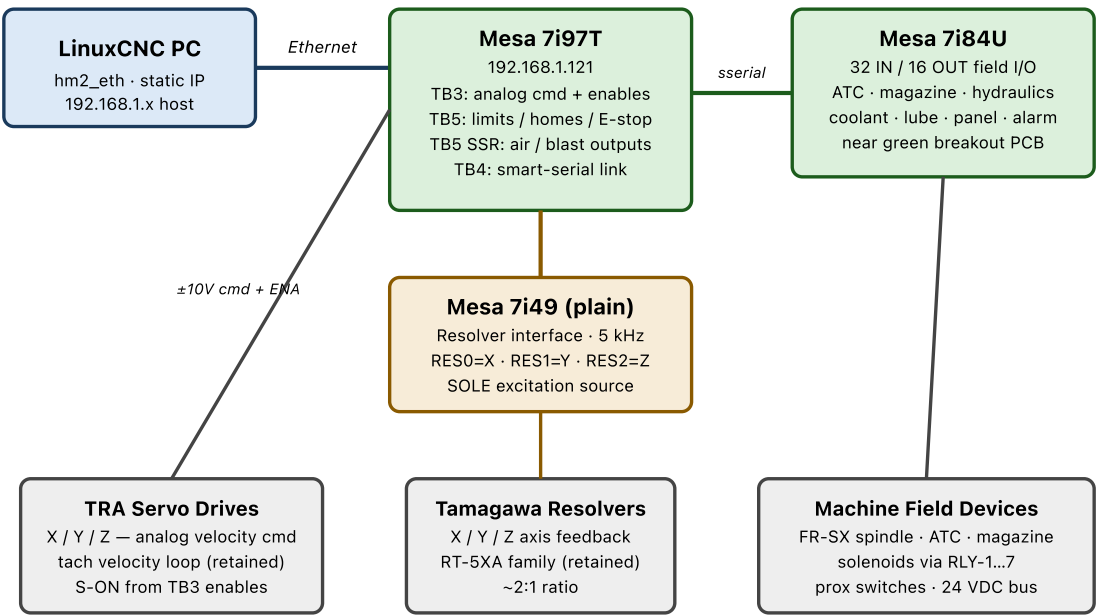


Figure 4-1. Control topology. The PC talks only to the 7i97T; the 7i84U hangs off smart-serial; the 7i49 provides resolver channels. All machine power electronics are retained.

§2 — 7i97T TB3: Servo Analog Commands & Enables

Note the channel order: **X = channel 0**, **Z = channel 1**, **Y = channel 2**.

TB3 pins	Signal	HAL net	Goes to	Status
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TB3.1 / TB3.2	ENA0– / ENA0+	x-enable	X servo S-ON	PLAN_VERIFY
TB3.3 / TB3.4	GND / AOUT0	x-vel-cmd	X drive analog cmd	PLAN_VERIFY
TB3.5 / TB3.6	ENA1– / ENA1+	z-enable	Z servo S-ON	PLAN_VERIFY
TB3.7 / TB3.8	GND / AOUT1	z-vel-cmd	Z drive analog cmd	PLAN_VERIFY
TB3.9 / TB3.10	ENA2– / ENA2+	y-enable	Y servo S-ON	PLAN_VERIFY
TB3.11 / TB3.12	GND / AOUT2	y-vel-cmd	Y drive analog cmd	PLAN_VERIFY
TB3.13 / TB3.14	ENA3– / ENA3+	spindle-enable	VFD FWD/RUN <i>candidate</i>	HOLD_CONFLICT
TB3.15 / TB3.16	GND / AOUT3	spindle-speed-cmd	FR-SX V-IN speed ref	PLAN_VERIFY

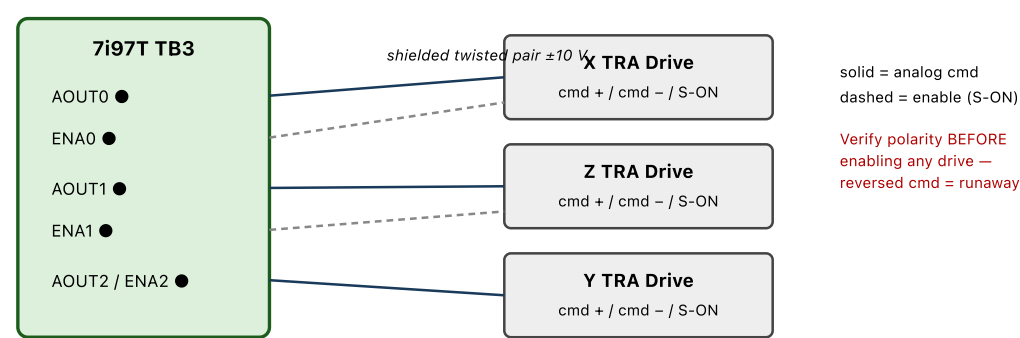


Figure 4-2. TB3 to TRA drives. Use shielded twisted pairs for analog; keep them away from contactor/motor power runs.

§3 — 7i49 Resolver Wiring

Channel	Axis	HAL net	Field device	Status
RES0	X	x-pos-fb	Tamagawa X resolver	PLAN_VERIFY
RES1	Y	y-pos-fb	Tamagawa Y resolver	PLAN_VERIFY
RES2	Z	z-pos-fb	Tamagawa Z resolver	PLAN_VERIFY

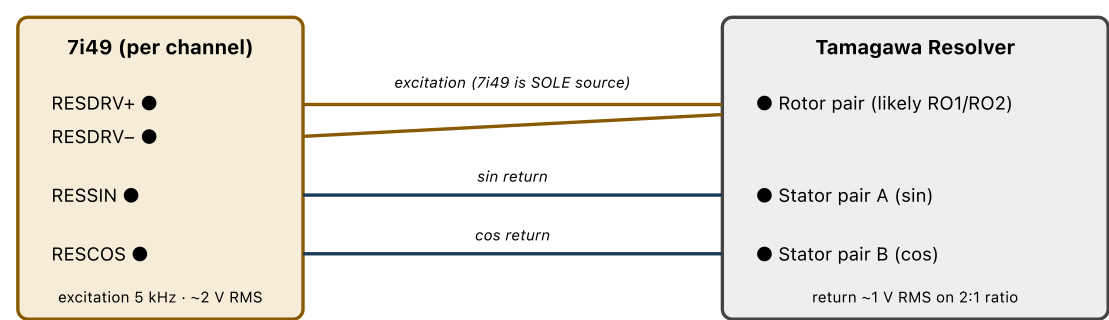


Figure 4-3. Resolver channel wiring. All pairs shielded; shield termination strategy to be finalized (Vol 3 §5). W2 half-drive jumper is a field-verification option only. Escalate to 7i49HV only if the return is far too weak at full drive.

§4 — 7i97T TB5: Limits, Homes, E-stop & SSR Outputs

TB5 pin	GPIO	HAL net	Field device	Status
TB5.1 IN0	gpio.008	limit-x-plus	X + overtravel	ACCEPTED_VERIFY
TB5.2 IN1	gpio.009	limit-x-minus	X – overtravel	ACCEPTED_VERIFY
TB5.3 IN2	gpio.010	limit-y-plus	Y + overtravel	ACCEPTED_VERIFY
TB5.4 IN3	gpio.011	limit-y-minus	Y – overtravel	ACCEPTED_VERIFY
TB5.5 IN4	gpio.012	limit-z-plus	Z + overtravel	ACCEPTED_VERIFY
TB5.6 IN5	gpio.013	limit-z-minus	Z – overtravel	ACCEPTED_VERIFY
TB5.7 IN6	gpio.014	home-x	X home switch	ACCEPTED_VERIFY
TB5.8 IN7	gpio.015	home-y	Y home switch	ACCEPTED_VERIFY
TB5.9 IN8	gpio.016	home-z	Z home switch	ACCEPTED_VERIFY
TB5.10 IN9	gpio.017	estop-ext	E-stop chain monitor	ACCEPTED_VERIFY
TB5.11	—	—	24 V field common (GND)	PLAN_VERIFY
TB5.12	—	—	+VFIELD 24 V supply	PLAN_VERIFY

TB5 SSR outputs (overflow relay drivers — coils are 100 VAC, always via interposing relay):

TB5 pins	GPIO	HAL net	Load	Status
TB5.13–14 SSR OUT0	gpio.020	air-blast	RLY-5 → SOL-62	PLAN_VERIFY
TB5.15–16 SSR OUT1	gpio.021	touch-sensor-blast	RLY-6 → SOL-35	PLAN_VERIFY
TB5.17–18 SSR OUT2	gpio.022	tap-coolant-blast	RLY-7 → SOL-61	PLAN_VERIFY
TB5.19–24 SSR OUT3–5	gpio.023–025	—	Spare — keep unassigned	SPARE

§5 — 7i97T TB4 ↔ 7i84U Smart-Serial Link

All rows **ACCEPTED**. Use absolute TB4 pin numbers. TX/RX labels are from each card's own perspective (7i97T TX → 7i84U RX).

7i97T TB4	Signal	7i84U RJ45 pin	Note
TB4.13	GND	4	ground A
TB4.14	GND	5	ground B — shield drain at 7i97T end only
TB4.15	RX+	6 (TX+)	verify continuity before power
TB4.16	RX–	3 (TX–)	verify continuity before power
TB4.17	TX+	2 (RX+)	verify continuity before power
TB4.18	TX–	1 (RX–)	verify continuity before power
TB4.19	+5V	7	serial power — confirm w/ 7i84U manual
TB4.20	+5V	8	serial power — confirm w/ 7i84U manual

§6 — Spindle Control: Mitsubishi FR-SX

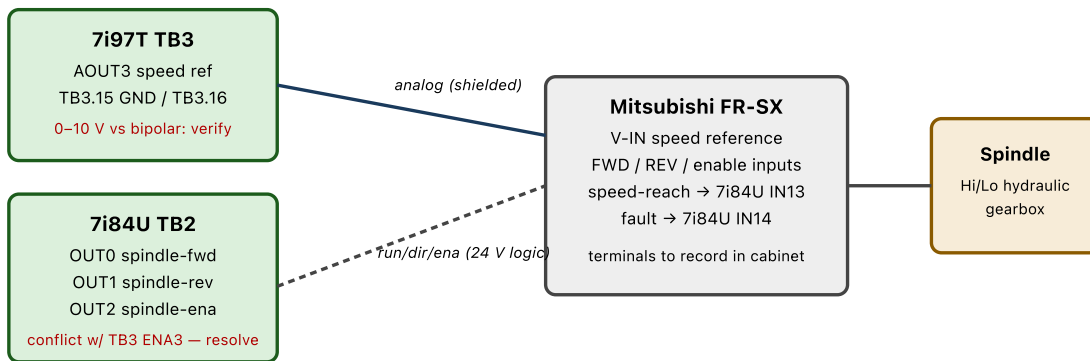


Figure 4-4. Spindle control. Gear shift: `gear-hi-sol` (OUT7 → RLY-1 → SOL-13) and `gear-lo-sol` (OUT8 → RLY-2 → SOL-12), confirmed by PRS-10 (IN17) / PRS-12 (IN18).

§7 — E-stop / Safety Chain Concept

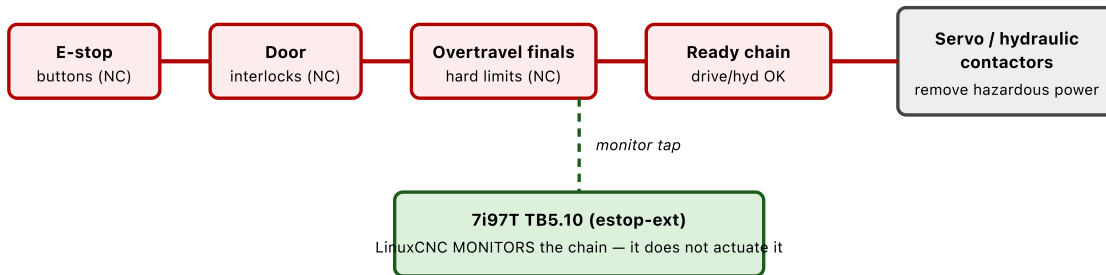


Figure 4-5. Series hardware chain drops the contactors directly. The Mesa input is a monitor tap only. Exact original chain wiring must be traced from schematic 41434WB and the cabinet before rewiring (Vol 3 §9).

§8 — 7i84U TB1: Complete Input Map (32 inputs)

Field power: TB1 pins 1/2 = bank A (+/–, feeds OUT0–7 & IN0–15); pins 3/4 = bank B (feeds OUT8–15 & IN16–31). Status is PLAN_VERIFY unless noted.

IN	HAL net	Field device / meaning
IN0	atc-y-zone	PRS-55 — Y axis in ATC zone
IN1	atc-z-zone	PRS-66 — Z axis in ATC zone
IN2	mag-tool-avail	PHS-181 — magazine tool available
IN3	spindle-tool-avail	PHS-182 — spindle tool available
IN4–IN9	—	Spare (keep for future field signals)
IN10	x-drive-fault	X TRA drive fault relay
IN11	y-drive-fault	Y TRA drive fault relay
IN12	z-drive-fault	Z TRA drive fault relay
IN13	spindle-at-speed	FR-SX speed-reach output
IN14	spindle-fault	FR-SX fault output
IN15	tool-clamped	PRS-9 — tool clamp confirm

IN16	tool-unclamped	PRS-8 — tool unclamp confirm
IN17	gear-hi-conf	PRS-10 — gear high confirm
IN18	gear-lo-conf	PRS-12 — gear low confirm
IN19	mag-bcd-bit0	PRS-21 — magazine BCD bit 0
IN20	mag-bcd-bit1	PRS-22 — magazine BCD bit 1
IN21	mag-bcd-bit2	PRS-23 — magazine BCD bit 2
IN22	mag-bcd-bit3	PRS-24 — magazine BCD bit 3
IN23	mag-bcd-bit4	PRS-25 — magazine BCD bit 4
IN24	door-interlock	Door interlock switch (state TBD after measurement)
IN25	lube-level	Lube level switch
IN26	coolant-level	Coolant level switch
IN27	hydraulic-ok	Sanwa SPS-8T-PC-20 pressure switch
IN28	cycle-start-pb	Cycle Start pushbutton (<i>optional panel reuse</i>)
IN29	feed-hold-pb	Feed Hold pushbutton (<i>optional panel reuse</i>)
IN30	single-block	Single Block selector (<i>optional panel reuse</i>)
IN31	servo-ready	Servo drives ready relay contact

Magazine BCD bits decode via `bcd2s` or a custom mux in HAL.

§9 — 7i84U TB2: Complete Output Map (16 outputs)

OUT	HAL net	Load	Notes
OUT0	spindle-fwd	FR-SX forward input	resolve vs TB3 ENA3 candidate
OUT1	spindle-rev	FR-SX reverse input	resolve vs TB3 ENA3 candidate
OUT2	spindle-ena	FR-SX enable input	resolve vs TB3 ENA3 candidate
OUT3– OUT5	—	Spare	drive S-ON handled at 7i97T TB3
OUT6	z-brake-rel	N1J-L2-201 Z brake release	flyback/suppression required
OUT7	gear-hi-sol	RLY-1 → SOL-13 (Fujikoshi hydraulic valve)	measure coil V before relay pick
OUT8	gear-lo-sol	RLY-2 → SOL-12 (Fujikoshi hydraulic valve)	measure coil V before relay pick
OUT9	tool-clamp-sol	RLY-3 → SOL-10	confirm single-valve vs dual-coil
OUT10	tool-unclamp-sol	RLY-4 → SOL-10	confirm single-valve vs dual-coil
OUT11	flood-coolant	Coolant pump relay	may need interposing relay
OUT12	lube-on	Lube pump relay	may need interposing relay
OUT13	atc-fwd	ATC motor forward relay	never energize before interlocks proven

OUT14	atc-rev	ATC motor reverse relay	never energize before interlocks proven
OUT15	alarm-out	Alarm light / horn (<i>optional</i>)	confirm load

§10 — Legend, Status Codes & Field Devices

Authority status codes (from `current_pin_authority.csv`):

Code	Meaning
ACCEPTED	Settled; land per table (still continuity-check first)
ACCEPTED_VERIFY	Assignment accepted; field device/normal state still to verify
PLAN_VERIFY	Engineered plan; trace in cabinet before landing wires
HOLD_CONFLICT	Two plans conflict — resolve before wiring either
RESERVED_VERIFY	Held for a possible device (e.g. spindle encoder)
OPTIONAL_VERIFY	Nice-to-have panel reuse
SPARE	Keep unassigned

Field-device prefix legend: PRS = proximity/position switch · PHS = photo/hall sensor · SOL = hydraulic/pneumatic solenoid · RLY = interposing relay (retrofit-added) · TRA = Mitsubishi servo drive family · FR-SX = Mitsubishi spindle drive.

Drawing cross-reference: machine-side power, motor, and original interlock wiring — Mazak schematic set **41434WB** (6/1984). Original PLC signal meanings — **YM2V39L** ladder manual and its extracted element list (385 signals).

End of Volume 4. Continue to Volume 5 — Maintenance & Troubleshooting.